

Appendix B

Heterogeneous DSR Availability Models and Dynamic Parameters

This appendix provides the detailed DSR availability models, dynamic parameter settings, and EV operating-mode classification procedure referenced in Section IV-B of the main manuscript.

The availability assessment first constructs type-level annual capacity profiles for PV, EV, and TCL. A predefined spatial allocation then maps the type-level profiles to the selected DSR buses. The nodal quantities with index i in the following expressions denote the final allocated capacities used in the planning model.

DSR Dynamic Parameters

The PV, EV, and TCL models are represented by the first-order lag transfer functions in Section II of the main manuscript. The parameter settings are listed in Table 1.

Table 1: DSR Transfer Function Parameters

Type	τ_k (s)	$\tau_{d,k}$ (s)	K_k
PV	0.2	0	25
EV	0.5	0.5	25
TCL	1.5	0.5	25

Photovoltaic DSR Capacity

Grid connected PV inverters participate in frequency regulation through de-loaded operation. The inverter output is capped at 95% of the forecast available power in this study. The remaining 5% provides upward regulation reserve. The upward regulation capacity at bus i under scenario s is:

$$P_{i,\text{PV},s}^{\text{avail,up}} = 0.05 \hat{P}_{i,\text{PV},s} \quad (1)$$

where $\hat{P}_{i,\text{PV},s}$ denotes the forecast available power of PV units at bus i under scenario s . The inverter operates at 95% of the forecast available power. The downward regulation capacity therefore equals the current operating power:

$$P_{i,\text{PV},s}^{\text{avail,dn}} = 0.95 \hat{P}_{i,\text{PV},s} \quad (2)$$

This de-loaded operating strategy is consistent with grid code requirements for active power reserve and primary frequency response. The 8760 hour annual output curve is obtained from irradiance data and inverter models. The forecast available power $\hat{P}_{i,\text{PV},s}$ is computed for each hour. It is then paired with the corresponding commitment scenario to form the available capacity sequence.

Electric Vehicle DSR Capacity

EV flexibility is assessed through Monte Carlo simulation of daily travel patterns. Each vehicle is assigned to one of four mutually exclusive operating modes, namely $S_{c,d}$, S_c , S_d , and S_0 . These modes represent charge and discharge, charge only, discharge only, and unavailable states. The classification depends on parking duration, state of charge requirements, grid connection status, and charging or discharging economics.

For each vehicle v , the simulation samples the initial state of charge $E_{\text{start}}^{(v)}$, the target state of charge $E_{\text{end},\tau}^{(v)}$, the arrival time $t_a^{(v)}$, the departure time $t_d^{(v)}$, the charging price $p_c^{(v)}$, and the discharging price $p_d^{(v)}$. These samples determine the available parking duration $\Delta t_p^{(v)} = t_d^{(v)} - t_a^{(v)}$, the required charging duration $T_{c,\text{need}}^{(v)}$, the possible discharging duration $T_{d,\text{possible}}^{(v)}$, and the operating mode.

After mode classification, the charging and discharging durations are mapped onto the hourly time axis. Individual power contributions are then aggregated across the fleet. This aggregation gives the theoretical maximum charging power $P_{i,\text{EV},s}^{\text{chg}}$ and the theoretical maximum discharging power $P_{i,\text{EV},s}^{\text{dis}}$ for each hour under scenario s . Discharging is defined as positive. Charging is defined as negative.

The upward regulation capacity represents the margin between the maximum discharging power and the actual operating power:

$$P_{i,\text{EV},s}^{\text{avail,up}} = \left(P_{i,\text{EV},s}^{\text{dis}} - P_{i,\text{EV},s}^{\text{op}} \right) \cdot \eta_{\text{agg}} \quad (3)$$

where $P_{i,\text{EV},s}^{\text{op}}$ denotes the actual operating power under scenario s , and η_{agg} is the aggregation efficiency factor. The downward regulation capacity represents the margin between the actual operating power and the maximum charging power:

$$P_{i,\text{EV},s}^{\text{avail,dn}} = \left(P_{i,\text{EV},s}^{\text{op}} - P_{i,\text{EV},s}^{\text{chg}} \right) \cdot \eta_{\text{agg}} \quad (4)$$

Random perturbations are added to the daily profiles to generate the full 8760 hour annual sequence. The capacity values are then matched to commitment scenarios.

EV Operating Mode Classification and Duration Calculation

The EV availability model uses the following operating-mode classification and duration calculation procedure.

Algorithm 1 EV Operating Mode Classification and Duration Calculation

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1: Input: Fleet size  $N_{EV}$ ; battery capacity  $C$ ; charge power  $P_c$ ; discharge power  $P_d$ ;
   charge efficiency  $\eta_c$ ; discharge efficiency  $\eta_d$ ; minimum SOC threshold  $E_{min}$ .
2: Output: Operating mode  $S_{ev}^{(v)}$ ; charge duration  $T_c^{(v)}$ ; discharge duration  $T_d^{(v)}$  for
   each vehicle  $v$ .
3: for  $v = 1$  to  $N_{EV}$  do
4:   Sample  $E_{start}^{(v)}, E_{end,\tau}^{(v)}, t_a^{(v)}, t_d^{(v)}, p_c^{(v)}, p_d^{(v)}$ .
5:    $\Delta t_p^{(v)} := t_d^{(v)} - t_a^{(v)}$ .
6:    $T_{c,need}^{(v)} := \max(E_{end,\tau}^{(v)} - E_{start}^{(v)}, 0) \cdot C / (P_c \eta_c)$ .
7:    $T_{d,possible}^{(v)} := \max(E_{start}^{(v)} - E_{min}, 0) \cdot C / (P_d \eta_d)$ .
8:    $F_{need,c}^{(v)} := [E_{start}^{(v)} < E_{end,\tau}^{(v)}]$ .
9:    $F_{need,d}^{(v)} := [E_{start}^{(v)} > E_{end,\tau}^{(v)}]$ .
10:   $F_{energy}^{(v)} := [E_{start}^{(v)} > E_{min}]$ .
11:   $F_{reach}^{(v)} := [E_{end,\tau}^{(v)} \geq E_{min}]$ .
12:   $F_{price}^{(v)} := [p_d^{(v)} > p_c^{(v)}]$ .
13:   $F_{time,cd}^{(v)} := [T_{d,possible}^{(v)} + T_{c,need}^{(v)} \leq \Delta t_p^{(v)}]$ .
14:   $F_{time,d}^{(v)} := [T_{d,possible}^{(v)} \leq \Delta t_p^{(v)}]$ .
15:   $F_{cd}^{(v)} := F_{need,c}^{(v)} \wedge F_{energy}^{(v)} \wedge F_{price}^{(v)} \wedge F_{time,cd}^{(v)} \wedge F_{reach}^{(v)}$ .
16:  if  $F_{cd}^{(v)}$  then
17:     $S_{ev}^{(v)} := S_{c,d}$ ;  $T_d^{(v)} := (E_{start}^{(v)} - E_{min})C / (P_d \eta_d)$ ;  $T_c^{(v)} := (E_{end,\tau}^{(v)} - E_{min})C / (P_c \eta_c)$ .
18:  else if  $F_{need,c}^{(v)} \wedge \neg F_{cd}^{(v)}$  then
19:     $S_{ev}^{(v)} := S_c$ ;  $T_c^{(v)} := T_{c,need}^{(v)}$ ;  $T_d^{(v)} := 0$ .
20:  else if  $F_{need,d}^{(v)} \wedge F_{price}^{(v)} \wedge F_{time,d}^{(v)} \wedge F_{energy}^{(v)}$  then
21:     $S_{ev}^{(v)} := S_d$ ;  $T_d^{(v)} := \min(E_{start}^{(v)} - E_{end,\tau}^{(v)}, E_{start}^{(v)} - E_{min})C / (P_d \eta_d)$ ;  $T_c^{(v)} := 0$ .
22:  else
23:     $S_{ev}^{(v)} := S_0$ ;  $T_c^{(v)} := 0$ ;  $T_d^{(v)} := 0$ .
24:  end if
25:  if  $T_c^{(v)} + T_d^{(v)} > \Delta t_p^{(v)}$  then
26:     $\lambda := \Delta t_p^{(v)} / (T_c^{(v)} + T_d^{(v)})$ ;  $T_c^{(v)} := \lambda T_c^{(v)}$ ;  $T_d^{(v)} := \lambda T_d^{(v)}$ .
27:  end if
28: end for
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Thermostatically Controlled Load DSR Capacity

TCL flexibility is derived from thermal comfort deadband dynamics. A device starts operation after the indoor temperature exceeds the deadband threshold around the setpoint. Over short intervals, the indoor temperature is approximated as quasi steady. The electrical power of a single device is linearly proportional to the outdoor temperature difference. The power is further adjusted by the coefficient of performance, a time of day factor, and an occupancy factor. The total power consumption is used to infer the number of operating devices n_{on} . The remaining devices are in the standby state $n_{standby}$.

The upward regulation capacity is obtained by switching off 40% of the

operating devices. The downward regulation capacity is obtained by switching on 40% of the standby devices. A device availability factor of 0.8 is applied as a conservative estimate based on operator practice. The aggregate adjustable capacities are:

$$P_{i,\text{TCL},s}^{\text{avail,up}} = 0.40 \times 0.8 \times n_{\text{on}} \cdot P_{i,\text{TCL}}^{\text{rated}} \quad (5)$$

$$P_{i,\text{TCL},s}^{\text{avail,dn}} = 0.40 \times 0.8 \times n_{\text{standby}} \cdot P_{i,\text{TCL}}^{\text{rated}} \quad (6)$$

where $P_{i,\text{TCL}}^{\text{rated}}$ denotes the rated power of a single device. The hourly values are computed over the full year and mapped to commitment scenarios.